PETER THE GREAT ST. PETERSBURG POLYTECHNIC UNIVERSITY

Vendor-Agnostic Multi-Agent LLM Workflow Orchestrator for Industrial Operations

A Thesis submitted in partial satisfaction of the requirements
for the degree Master of Science

in

Intelligent Systems

by

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~~The Thesis of Vendor-Agnostic Multi-Agent LLM Workflow Orchestrator for Industrial Operationsis approved, and it is acceptable in quality and form for publication on microfilm and electronically.~~

Peter The Great St. Petersburg Polytechnic University

2025

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ABSTRACT OF THE THESIS

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by

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Master of Science in Intelligent Systems

Peter The Great St. Petersburg Polytechnic University, 2025

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Abstract: Modern industrial systems increasingly rely on distributed automation across heterogeneous platforms, creating a growing demand for intelligent coordination mechanisms that go beyond traditional centralized control architectures. Recent advances in Large Language Models (LLMs) have demonstrated strong reasoning and contextual understanding capabilities, motivating their integration into multi-agent systems for industrial workflow orchestration.
This thesis presents a systematic analytical review of LLM-based multi-agent systems for industrial operations. The study examines representative research solutions (ToolNet, AFLOW, SOAN, and LLM4IAS) alongside industrial patent-based systems (ABB EP3929679A1 and Phaidra US20250021061A1), drawing on domestic and foreign scientific sources published between 2020 and 2025. The analysis is conducted using a structured qualitative framework covering architectural design, coordination mechanisms, deployment readiness, safety considerations, and interoperability.

The review reveals a clear structural divide between research-oriented and industrial solutions. Research systems demonstrate advanced cognitive capabilities, including LLM-driven reasoning, flexible workflow modeling, and decentralized agent coordination, but lack industrial deployability, safety certification, and real-time guarantees. In contrast, industrial solutions emphasize robustness, safety, and operational readiness, yet deliberately constrain intelligence, openness, and cross-vendor interoperability, resulting in strong vendor lock-in. A qualitative comparison across eight key characteristics shows that none of the analyzed systems simultaneously satisfies all identified requirements, with vendor-agnostic interoperability emerging as a universal gap.

The outcomes of this review define explicit design requirements and architectural principles for a vendor-agnostic multi-agent LLM workflow orchestrator, serving as a concrete analytical foundation for the subsequent design and implementation stages of the thesis.

INTRODUCTION

1.1 Background and Motivation

Over the past decade, industrial systems have undergone a rapid digital transformation driven by advances in automation, data analytics, and artificial intelligence. Modern industrial environments increasingly rely on heterogeneous software platforms, intelligent devices, and distributed control systems provided by multiple vendors. While this technological diversity enables flexibility and scalability, it also introduces significant challenges related to interoperability, coordination, and system-level intelligence.

At the same time, the recent emergence of Large Language Models (LLMs) has fundamentally changed the landscape of artificial intelligence. Unlike traditional machine learning models, LLMs demonstrate advanced reasoning, contextual understanding, and decision-support capabilities. These properties have stimulated growing interest in their use as intelligent agents capable of coordinating complex workflows, interacting with tools, and supporting human operators in decision-making processes.

In parallel, multi-agent systems (MAS) have been extensively studied as a paradigm for modeling distributed intelligence in complex environments. By decomposing tasks among multiple autonomous agents, MAS enable modularity, adaptability, and scalability. However, most existing industrial applications of multi-agent systems remain tightly coupled to specific vendors, platforms, or predefined workflows, limiting their applicability in real-world heterogeneous industrial settings.

The convergence of LLM-based reasoning and multi-agent architectures creates new opportunities for addressing long-standing industrial challenges, particularly in workflow orchestration, cross-system coordination, and intelligent decision support. Nevertheless, current

research and industrial solutions remain fragmented, lacking a unified analytical understanding of existing approaches and their limitations.

1.2 Research Topic and Scope

This analytical review focuses on existing scientific and practical solutions related to LLM-based multi-agent systems for industrial applications, with particular emphasis on workflow orchestration and coordination across heterogeneous industrial environments.

The scope of this review includes:

- LLM-based autonomous and semi-autonomous agents;
- Multi-agent architectures applied to industrial decision-support tasks;
- Workflow orchestration mechanisms and tool coordination strategies;
- Solutions addressing interoperability and vendor heterogeneity in industrial systems.

The review does **not** aim to propose a new system architecture or implementation at this stage. Instead, its primary objective is to analyze existing approaches, identify key research trends, and reveal unresolved gaps that motivate further scientific investigation.

1.3 Keywords and Search Strategy

To ensure comprehensive coverage of the subject area, a structured keyword-based search strategy was adopted. The following keywords and their semantic variations were used:
Multi-agent systems, *LLM-based agents*, *large language models in industry*, *Agentic AI*, *Workflow orchestration*, *Industrial automation*, *Vendor-agnostic systems*, *Industrial interoperability*, *Tool orchestration*, and *Autonomous AI agents*

Creative keyword reformulations and cross-domain translations were applied to capture relevant publications from both computer science and industrial engineering perspectives.

1.4 Information Sources and Databases

The literature search was conducted using the following scientific databases and research platforms:

- Google Scholar
- Semantic Scholar
- ResearchGate
- CyberLeninka
- RSCI (eLibrary)
- The Lens

These sources were selected to ensure access to both international and domestic scientific publications, peer-reviewed journal articles, conference proceedings, and relevant open-access research outputs. The search was limited to publications from 2022 to 2026 in order to reflect the most recent developments in the field.

1.5 Structure of the Analytical Review

This analytical review is organized as follows. Section 2 presents the key areas of research and development related to LLM-based multi-agent systems in industrial contexts. Section 3 analyzes existing scientific solutions and identifies dominant architectural and methodological trends. Section 4 focuses on finished products and practical systems derived from scientific research. Section 5 introduces qualitative characteristics used for comparison. Section 6 provides a comparative analysis of competitive solutions. Section 7 discusses the identified gaps and limitations of existing approaches. Finally, Section 8 summarizes the main findings and outlines directions for further research.

2 Key Areas of Research and Development

2.1 LLM-Based Multi-Agent Systems in Industrial Environments

Recent research highlights a growing interest in the integration of Large Language Models (LLMs) into multi-agent systems (MAS) for industrial environments. In these studies, LLMs are employed as high-level reasoning components that enable agents to interpret complex instructions, reason over heterogeneous data sources, and dynamically adapt their behavior in response to changing operational conditions.

Several works demonstrate the applicability of this paradigm to manufacturing shopfloors and industrial maintenance scenarios, reporting improvements in adaptability, coordination efficiency, and deployment flexibility *(e.g., P1, P5, P7).* A common architectural pattern involves decomposing industrial tasks among multiple autonomous agents that coordinate through negotiation or message-based interaction. However, despite their promising results, many of these solutions remain tightly coupled to particular industrial tasks or domains, which limits their transferability to broader industrial contexts and heterogeneous environments.

2.2 Agentic AI for Industrial Decision Support and Automation

A second major research direction focuses on the use of agentic AI systems for industrial decision support and partial automation. In this context, LLM-based agents are designed to translate high-level human intents into operational decisions, assist in diagnostics, and generate explainable recommendations for industrial operators.

Recent studies apply this approach to predictive maintenance, fault diagnosis, and intent-based industrial automation *(e.g., P2, P3, P7)*. A recurring design principle across these works is the explicit inclusion of human-in-the-loop mechanisms to ensure safety, accountability, and trust. While such mechanisms enhance system reliability in safety-critical environments, they also reveal

unresolved challenges related to defining appropriate levels of agent autonomy and responsibility distribution between human operators and intelligent agents.

2.3 Workflow Orchestration and Tool-Based Coordination

Workflow orchestration has emerged as a critical layer for coordinating complex industrial processes involving multiple tools, services, and information sources. Recent research increasingly adopts graph-based abstractions to model workflows, where tasks or tools are represented as nodes and their dependencies as edges.

Several studies introduce formal representations in which tools are modeled as nodes within explicit task or tool graphs, enabling structured reasoning and dynamic workflow adaptation *(e.g., P11, P12, P10).* These representations allow LLM-based agents to plan, execute, and modify workflows in response to contextual changes. Nevertheless, most existing orchestration approaches are evaluated in isolation and lack seamless integration with broader industrial interoperability requirements, which limits their applicability in real-world multi-vendor environments.

2.4 Centralized vs. Decentralized Multi-Agent Architectures

The architectural organization of multi-agent systems represents a fundamental design choice in industrial applications. Centralized architectures rely on a single coordinating entity responsible for global task allocation and decision-making, whereas decentralized architectures distribute control among multiple autonomous agents.

The reviewed literature indicates that centralized approaches simplify coordination and system monitoring but introduce scalability limitations and single points of failure *(e.g., P4, P6).* In contrast, decentralized and hybrid architectures improve robustness and scalability by distributing decision-making across agents *(e.g., P1, P12, P14).* Despite extensive discussion of

these trade-offs, systematic experimental comparisons between centralized and decentralized architectures remain limited, leaving open questions regarding their suitability for different industrial scenarios.

2.5 Industrial Interoperability and Vendor Heterogeneity

Industrial interoperability and vendor heterogeneity constitute persistent challenges across all examined research directions. Modern industrial environments typically rely on equipment and software systems provided by multiple vendors, each exposing proprietary interfaces and data models.

Several studies emphasize the role of open standards and standardized communication mechanisms in mitigating vendor lock-in and enabling cross-system coordination *(e.g., P4, P10, P20).* In particular, standards such as IEC 61499 and interoperability protocols such as OPC UA are frequently cited as key enablers of vendor-independent system integration. However, most LLM-based agent systems either assume homogeneous environments or address interoperability only at a conceptual level, leaving the problem of vendor-agnostic orchestration largely unresolved in practice.

Overall, the reviewed literature demonstrates significant progress across multiple research directions, including LLM-based agent reasoning, industrial decision support, workflow orchestration, architectural design, and interoperability. However, these directions are often explored independently. The lack of integrated solutions that simultaneously address agent coordination, structured workflow orchestration, architectural robustness, and vendor-agnostic interoperability highlights a clear gap in the current state of the art and motivates further analytical and experimental investigation.

3. Analysis of Existing Scientific Solutions

To provide a systematic understanding of the current state of the art, this section analyzes representative scientific solutions addressing LLM-based multi-agent systems in industrial contexts. Rather than summarizing individual papers, the analysis groups existing solutions into coherent categories based on their functional focus, architectural design, and evaluation methodology. This approach enables the identification of common strengths, recurring limitations, and unresolved challenges across the literature.

3.1 LLM-Based Multi-Agent Solutions for Industrial Tasks

A substantial body of recent research proposes LLM-based multi-agent solutions tailored to specific industrial tasks, including manufacturing scheduling, shopfloor coordination, and maintenance planning. In these systems, multiple autonomous agents collaborate by exchanging messages or negotiating decisions under predefined task constraints.

Empirical results reported in the literature indicate strong performance within narrowly defined industrial scenarios. For example, ***P1*** demonstrates effective coordination in two-step and four-step machining workflows, achieving success rates of **100%** and **86%**, respectively. Similarly, ***P5*** reports a reduced makespan in flexible job-shop scheduling within a single milling shopfloor through agent-based negotiation mechanisms. These results highlight the potential of LLM-enabled agents to improve efficiency and adaptability when task boundaries are clearly specified.

However, a recurring limitation across these solutions is their tight coupling to particular industrial tasks or environments. Agent behaviors, communication protocols, and coordination logic are often designed around fixed workflows, which constrains reuse when tasks, tools, or operational contexts change. This task-specific coupling limits the scalability of such solutions

beyond their original experimental settings and motivates the need for more generalizable coordination mechanisms.

3.2 Agentic Decision-Support Systems with Human-in-the-Loop

Another prominent class of solutions focuses on agentic AI systems designed to support human decision-making rather than fully automate industrial processes. In these approaches, LLM-based agents interpret high-level intents, analyze system states, and provide recommendations, explanations, or diagnostic insights to human operators.

Studies such as ***P2*** and ***P3*** apply this paradigm to intent-based industrial automation and predictive maintenance, demonstrating that LLM-based agents can effectively bridge the gap between business-level objectives and operational-level actions. A defining characteristic of these systems is the explicit inclusion of human-in-the-loop mechanisms to ensure safety and accountability, which is particularly important in safety-critical industrial environments.

Nevertheless, many human-in-the-loop designs lack formal models for balancing autonomy and oversight. For instance, while ***P2*** and ***P3*** incorporate intent decomposition and LLM fine-tuning, neither defines quantitative boundaries for when agents should defer decisions to human operators. Moreover, these works do not provide principled frameworks for allocating decision rights across agent–human boundaries in multi-step industrial workflows. As a result, the scalability of such systems beyond isolated decision-support tasks remains uncertain.

3.3 Workflow-Oriented Agent Systems and Tool Coordination

A growing research direction addresses the coordination of multiple tools and services through workflow-oriented agent systems. These solutions emphasize structured task decomposition and execution control, enabling agents to interact with heterogeneous tools in a goal-driven manner.

Recent studies increasingly rely on explicit graph-based workflow representations, where tools or tasks are modeled as nodes and execution dependencies as edges *(e.g., **P11, P12, P10**)*. For example, ***P11*** (ToolNet) evaluates tool graphs on abstract benchmarks such as ToolBench and APIBank, involving up to **3,992 APIs**, demonstrating the scalability of graph-based tool selection. However, this evaluation does not consider industrial communication protocols such as OPC UA or Modbus. Similarly, ***P12*** (AFLOW) optimizes agent workflows for coding and reasoning tasks but does not address real-time constraints, safety interlocks, or execution determinism typical of industrial control systems.

Consequently, while these solutions demonstrate conceptual maturity in workflow reasoning, they are primarily evaluated in isolated experimental settings. Integration with real industrial ecosystems characterized by vendor heterogeneity, legacy systems, and strict operational constraints remains limited, highlighting a gap between workflow-oriented research and industrial deployment requirements.

3.4 Architectural Organization of Existing Solutions

The architectural organization of multi-agent systems varies significantly across the analyzed solutions, ranging from centralized coordination models to decentralized and hybrid agent networks. Centralized architectures simplify coordination and monitoring by relying on a single control entity *(e.g., **P4, P6**)*, but they introduce scalability limitations and single points of failure.

Decentralized and hybrid architectures distribute decision-making across agents, improving robustness and adaptability (*e.g., **P1, P12, P14**).* Notably, ***P14*** (SOAN) provides one of the few quantitative comparisons between architectural paradigms, showing that structure-aware, self-organizing agent networks maintain pass@1 accuracy above 90% on complex nested

workflows. In contrast, flat centralized baselines such as AutoGen and CAMEL exhibit sharp performance degradation as workflow depth and length increase.

Despite this contribution, even SOAN evaluates only enterprise business workflows and does not consider industrial control scenarios involving real-time constraints, safety interlocks, or vendor-specific PLC interfaces. Overall, systematic experimental comparisons across architectures in realistic industrial settings remain scarce.

3.5 Interoperability-Oriented Industrial Agent Solutions

A smaller subset of the literature explicitly addresses interoperability and vendor heterogeneity in industrial agent systems. These works emphasize the use of open standards and standardized communication protocols to enable cross-system coordination.

For example, ***P4*** demonstrates IEC 61499-based distributed control architectures while highlighting tool immaturity and performance challenges. ***P10*** achieves LLM-based industrial control via OPC UA, but remains tied to a single automation setup. ***P20*** introduces the Model Context Protocol (MCP) to standardize REST API integration, yet does not address industrial protocols or multi-vendor PLC environments.

Consequently, agent coordination and workflow orchestration are rarely designed to operate seamlessly across multiple vendors and platforms. This observation underscores a critical gap: vendor-agnostic interoperability is typically treated as an auxiliary feature rather than a primary design objective. Addressing this gap is essential for enabling scalable and transferable industrial multi-agent systems.

In summary, existing scientific solutions demonstrate substantial progress in LLM-based agent reasoning, decision support, workflow orchestration, and architectural design. However, these advances are fragmented across task-specific, workflow-oriented, or interoperability-focused

approaches. Few solutions integrate these dimensions within a unified framework capable of operating under realistic industrial constraints. This fragmentation motivates the need for a comparative analysis based on well-defined qualitative characteristics and provides a clear foundation for identifying research gaps in the subsequent sections.

4. Finished Products and Practical Systems

4.1 Selection Criteria and Terminology Clarification

Before analyzing concrete systems, it is necessary to clarify the terminology used in this section. In the context of this analytical review, the term finished product does not imply a fully commercialized industrial solution. Instead, it refers to systems that have progressed beyond abstract methods or isolated algorithms and demonstrate a complete system-level implementation, including a defined architecture, execution logic, and empirical evaluation.

Accordingly, this section distinguishes between research-derived finished systems and purely conceptual approaches. A system is considered a finished product if it satisfies the following criteria:

(1) it implements an end-to-end workflow or control pipeline rather than a single algorithmic component;

(2) it provides a clear architectural decomposition of system components;

(3) it has been empirically evaluated using benchmarks, datasets, or physical testbeds; and

(4) it is usable by identifiable groups of users, such as researchers, developers, or industrial partners.

Based on these criteria, four representative systems derived from recent scientific articles were selected for detailed analysis. These systems reflect different design philosophies and application contexts, while collectively illustrating the current capabilities and limitations of practical multi-agent and LLM-based systems.

4.2 Research-Derived Finished Systems (Article-Based)

4.2.1 ToolNet: Graph-Based Tool Orchestration System

ToolNet is a tool orchestration system developed through a research collaboration involving Northwestern University, North Carolina State University, and Microsoft Research Asia, led by Xukun Liu et al. (2024). The development team includes researchers such as Zhiyuan Peng, Xiaoyuan Yi, and Xing Xie from Microsoft Research. The system was proposed in response to the growing difficulty of scaling tool-augmented large language models beyond small collections of tools.

From a technical perspective, ToolNet introduces a graph-based orchestration architecture in which tools are explicitly modeled as a weighted directed graph ($G = (V, E)$). Each node represents an individual tool or API, annotated with metadata such as input/output schemas and natural language descriptions. Directed edges encode transition likelihoods between tools, which are learned from historical execution trajectories. An LLM-based evaluator module scores tool invocations and dynamically updates edge weights based on execution success.

The motivation behind ToolNet arises from a fundamental limitation observed in conventional in-context tool learning: as the number of available tools increases beyond a few hundred, prompt-based selection becomes unreliable due to token budget constraints and hallucinated tool calls. ToolNet addresses this issue by constraining the agent's search space to local neighborhoods within the tool graph, enabling scalable navigation of large tool ecosystems. The system has been empirically evaluated on ToolBench and APIBank, covering up to 3,992 APIs and more than 16,000 tool-use trajectories.

ToolNet is primarily used by academic researchers and developers working on tool-augmented LLM systems and agent frameworks. Its open-source implementation, written in Python and built on top of PyTorch and graph-processing libraries, has been adopted as a reference system for evaluating large-scale tool orchestration strategies. However, ToolNet does not support

industrial communication protocols such as OPC UA, Modbus, or PROFINET, nor does it provide real-time execution guarantees or safety interlocks. As a result, while ToolNet constitutes a mature and well-engineered orchestration product in the general AI domain, its applicability to industrial automation remains limited.

4.2.2 AFLOW: Agentic Workflow Optimization Framework

AFLOW is an agentic workflow optimization framework developed by a multi-institutional research team led by Jiayi Zhang (Peking University), with contributions from researchers affiliated with Alibaba DAMO Academy, Tsinghua University, and the University of Montreal. The system was introduced to address limitations in manually designed agent workflows, particularly in complex multi-step reasoning and generation tasks.

Technically, AFLOW formulates workflow optimization as a search problem over structured execution graphs. Workflows are represented as directed graphs in which nodes correspond to LLM invocation steps, parameterized by model choice, prompts, decoding parameters, and output formats, while edges represent control flow constructs such as branching, loops, and conditional execution. The framework employs Monte Carlo Tree Search (MCTS) to explore the space of possible workflow configurations.

During execution, AFLOW iteratively expands candidate workflows by modifying graph structures through LLM-driven code transformations. Each candidate workflow is evaluated on validation datasets, and performance metrics are propagated back through the search tree using the Upper Confidence Bounds for Trees (UCT) algorithm. This process allows AFLOW to automatically discover workflow structures that outperform manually engineered baselines.

AFLOW is distributed as an open-source framework and has been evaluated on a wide range of benchmark tasks, including HumanEval, MBPP, GSM8K, MATH, HotpotQA, and

DROP. Experimental results show that optimized workflows generated by AFLOW can outperform hand-crafted pipelines by an average margin of 5–6%, while enabling smaller models to achieve performance comparable to much larger LLMs at significantly reduced inference cost.

The primary users of AFLOW are researchers and advanced practitioners studying agentic workflows and automated pipeline design. Despite its strong technical foundations and comprehensive evaluation, AFLOW does not address industrial execution environments. It lacks integration with PLCs, SCADA systems, or real-time control loops, and its optimization objectives focus on task accuracy rather than industrial metrics such as latency, safety, or operational efficiency. Consequently, AFLOW represents a finished and powerful research product, but one that remains disconnected from industrial automation contexts.

4.2.3 SOAN: Self-Organizing Agent Networks

SOAN (Self-Organizing Agent Networks) is a finished multi-agent system developed by a research team from Beihang University in collaboration with the Beijing Academy of Artificial Intelligence (BAAI). The system was introduced by Yiming Xiong et al. and published at AAAI 2026, one of the leading international conferences in artificial intelligence. The development team includes Jian Wang and Bing Li (Beihang University), as well as Yuhan Zhu and Yuqi Zhao (BAAI).

The primary motivation behind SOAN is to address the scalability limitations of centralized agent orchestration frameworks when confronted with increasingly complex and deeply nested workflows. Rather than relying on a single coordinator or flat agent structure, SOAN proposes a self-organizing coordination mechanism in which agent networks dynamically restructure themselves according to task complexity and execution context.

From a technical standpoint, each agent in SOAN is defined as a structured entity characterized by a goal, an internal process, an associated toolset, and contextual activation constraints. Complex workflows are first decomposed into atomic goal–process units. These units are then analyzed to detect recurring structural patterns, such as sequential composition, branching, and nested subflows. Identified patterns are encapsulated into higher-level agents, allowing the system to construct hierarchical agent networks that mirror the structural properties of the workflow itself.

A key feature of SOAN is its dynamic life-value maintenance mechanism, which continuously evaluates the utility of agents based on their contribution to successful task execution. Agents with low utility are pruned, while high-utility agents are retained and reused. This enables the system to adapt over time and maintain efficiency as task distributions evolve.

SOAN is evaluated on gFlowQA, a dataset containing approximately 8,000 enterprise workflows spanning domains such as customer service, IT operations, human resources, and supply chain management. The workflows vary in complexity from simple linear sequences to deeply nested structures with multiple branching levels. Experimental results show that SOAN achieves pass@1 accuracy of over 95% for shallow workflows and maintains performance above 90% for moderately nested workflows, significantly outperforming centralized baselines such as AutoGen, CAMEL, and MetaGPT as workflow depth increases.

The system implementation and experimental artifacts of SOAN are publicly available through the project repository, enabling reproducibility and further research. The availability of code, datasets, and evaluation scripts positions SOAN not only as a conceptual contribution but as a reusable research-grade system that can be extended and empirically validated by other research groups.

The system is primarily used by enterprise workflow automation researchers and academic groups studying large-scale multi-agent coordination. However, SOAN does not target industrial control scenarios. Its workflows represent business processes rather than physical control sequences, and the system does not address real-time execution constraints, safety interlocks, or vendor-specific industrial interfaces. Consequently, while SOAN constitutes a mature and well-evaluated finished product for decentralized agent coordination, its applicability to industrial automation remains indirect.

4.2.4 LLM4IAS: OPC UA–Based Industrial LLM Control System

LLM4IAS (Large Language Model for Industrial Automation Systems) is a finished and operational system developed by the Institute of Industrial Automation and Software Engineering (IAS) at the University of Stuttgart, Germany. The system was designed and implemented by Yuchen Xia under the supervision of Prof. Dr.-Ing. Michael Weyrich, with contributions from Nasser Jazdi, Jize Zhang, and Chaitanya Shah. The system was presented at IEEE ETFA 2025, a leading conference in industrial automation and factory systems.

The motivation behind LLM4IAS is to bridge the gap between high-level AI-driven reasoning and low-level industrial control systems. Unlike purely conceptual agent frameworks, LLM4IAS directly interfaces with physical industrial equipment through standardized communication protocols, demonstrating the feasibility of LLM-based control in real industrial settings.

Architecturally, LLM4IAS consists of four tightly integrated layers. The interoperability layer connects programmable logic controllers (PLCs), sensors, and actuators via OPC UA, supporting industrial Ethernet protocols such as PROFINET. The digital twin layer maintains a multi-level semantic representation of the physical system, converting low-level sensor data into

structured events that can be consumed by higher-level agents. The multi-agent control layer includes a manager agent responsible for global task planning, several operator agents controlling individual modules, and a summarization agent that generates execution reports. Finally, the LLM engine performs reasoning and decision-making using models such as GPT-4o and fine-tuned open-source alternatives.

The system has been validated on a physical industrial testbed located at the IAS laboratory in Stuttgart. The testbed includes ABB industrial robots, conveyor systems, vision inspection stations, and Siemens distributed I/O modules, executing a multi-step assembly process with multiple product variants. Experimental evaluation on 100 test cases demonstrates command correctness rates exceeding 90%, with average response times of approximately 1.8 seconds, which are acceptable for supervisory-level control but insufficient for hard real-time applications.

LLM4IAS is primarily used in research-oriented industrial automation environments and collaborative projects with industrial partners, including exploratory evaluations with Siemens AG and Bosch Rexroth. The system is available as an open-source research prototype, including source code, datasets, and containerized deployment artifacts. The complete system implementation, including Docker containers for the digital twin environment and evaluation scripts, is available through the project repository (e.g., *GitHub*), facilitating reproducibility and controlled experimentation across similar industrial testbeds.

Despite its strong industrial grounding, LLM4IAS remains limited in scope. It has been tested on a single automation setup and does not support orchestration across multiple vendors or production lines. Moreover, the system is not safety-certified and does not meet stringent real-time requirements. Nevertheless, LLM4IAS represents one of the most concrete and technically grounded examples of an LLM-based multi-agent control system in an industrial context.

Table 4. 1 Comparative Overview of Research-Derived Finished Systems

Dimension	ToolNet	AFLOW	SOAN	LLM4IAS
Institution	Northwestern Univ. + MSRA	Peking Univ. + Alibaba DAMO	Beihang Univ. + BAAI	Univ. of Stuttgart (IAS)
Year	2024	2025	2026	2025
Primary Focus	Tool orchestration	Workflow optimization	Self-organizing agents	Industrial integration
Architecture	Weighted directed tool graph	MCTS over workflow graphs	Hierarchical agent networks	Four-layer industrial architecture
Evaluation Environment	ToolBench, APIBank (3992 APIs)	HumanEval, MBPP, GSM8K, MATH	gFlowQA (≈8000 workflows)	Physical industrial testbed
Performance Highlights	Token-efficient tool selection	+5–6% over handcrafted workflows	>95% shallow, >90% nested	>90% command correctness
Open Source	Yes	Yes	Yes	Yes
Industrial Protocols	No	No	No	OPC UA
Real-Time Capability	No	No	No	Supervisory level (~1.8s)
Safety Interlocks	No	No	No	Not certified
Multi-Vendor Support	No	No	No	Single setup
Primary Users	Researchers, developers	Researchers, ML engineers	Enterprise workflow researchers	Industrial researchers, automation partners

4.3 Positioning of Research-Derived Finished Systems

The four analyzed systems represent distinct but complementary approaches to operationalizing agent-based and LLM-driven intelligence. ToolNet demonstrates scalable tool orchestration through graph-based representations, AFLOW provides systematic workflow optimization through search-based methods, SOAN illustrates the advantages of decentralized self-organizing coordination, and LLM4IAS offers concrete integration with industrial automation infrastructures.

Despite their maturity, these systems expose a shared limitation: none simultaneously integrates scalable agent coordination, structured workflow orchestration, and vendor-agnostic industrial interoperability within a unified architecture. This observation confirms that current research-derived finished systems, while technically advanced, remain fragmented across domains and application contexts. The identification of this gap provides a direct foundation for defining qualitative comparison characteristics and motivates the subsequent comparative analysis.

4.4 Patent-Based Industrial Solutions (Improved Draft)

4.4.1 ABB Patent EP3929679A1 - Modular Plant Orchestration via a Process Orchestration Layer (POL)

Patent identification.

This subsection examines the ABB patent EP3929679A1, attributed to Katharina Stark and Mario Hoernicke, with priority 2020-06-23 and publication 2021-12-29 (as listed in the patent record). The patent focuses on engineering-time orchestration for modular process automation, aiming to reduce the integration effort required to assemble a complete plant from reusable modules.

- What the patented solution is

At its core, EP3929679A1 proposes an orchestration-centric engineering workflow in which a plant is not engineered as one monolithic control program, but rather composed from independent automation modules. A dedicated layer referred to as the Process Orchestration Layer (POL) coordinates module behavior at the plant level. This is not presented as an abstract academic method; it is framed as a practical engineering approach intended to be embedded into industrial toolchains for modular automation.

- **How it works (technical core from the patent logic)**

The patent's main technical idea can be summarized as a pipeline with three engineering blocks:

1. Project-independent module engineering

   Each module is engineered as a self-contained automation unit with:

- clearly defined interfaces (signals, events, parameters),
- an internal control function,
- and constraints relevant to safe coordination with other modules.

2. Automatic generation / compilation of orchestration artifacts

   A composition procedure generates plant-level coordination artifacts from module descriptions. Conceptually, the orchestration process resembles building a plant orchestration graph where:

- nodes correspond to modules (or module functions),
- edges represent orchestration dependencies (sequencing, synchronization, resource coupling).

3. Project-dependent engineering of POL and deployment

   Finally, POL logic is integrated into the actual plant automation project, so that orchestration is executable in the runtime environment.

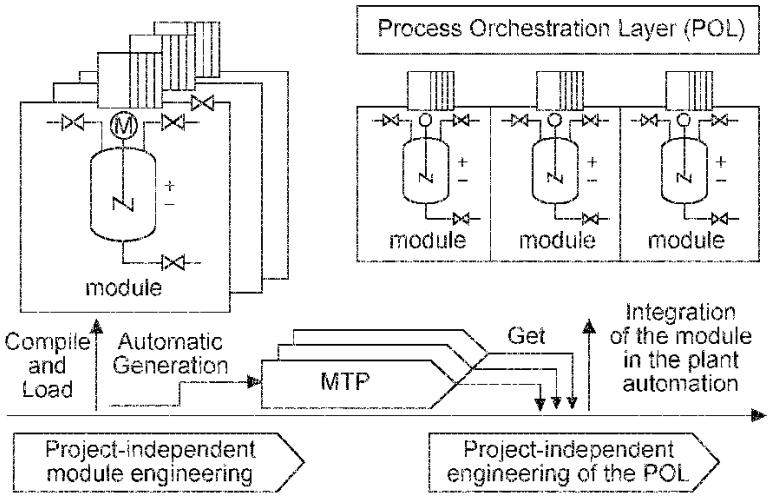


Fig. 1

Figure 4. 1 the Process Orchestration Layer (POL) composes multiple

- **What Figure 1 tells us (your provided drawing)**

Figure 1 visually communicates the "engineering-to-deployment" flow:

- On the left, multiple modules are engineered independently ("project-independent module engineering").
- In the middle, an Automatic Generation step compiles orchestration artifacts (the figure explicitly shows "Compile and Load" and "Automatic Generation").
- On the right, the Process Orchestration Layer (POL) coordinates the modules and enables "Integration of the module in the plant automation," which is the practical outcome: faster, more systematic commissioning of modular plants.

In other words, the figure supports a key claim: orchestration becomes a first-class engineering object, rather than being hidden inside ad-hoc PLC glue code.

- **Why it was patented (industrial motivation)**

The commercial rationale is straightforward: in large modular plants, the most expensive part is often not the module code itself, but the integration effort the sequencing logic, interlocks, coordination, and reconfiguration. ABB's patent protects a structured orchestration approach that:

- lowers integration time (by standardizing module composition), improves reuse (modules can be reused across projects), and makes plant-level behavior more maintainable through a dedicated orchestration layer.

- **Who uses it (realistic target users)**

While patents rarely list customer names, the intended users are concrete and recognizable in industrial practice:

- process automation engineers designing modular plants and procedures,
- system integrators responsible for commissioning and integration,
- vendor engineering teams who maintain module libraries and orchestration templates.

The value proposition is strongest in settings where modular automation is a standard approach (e.g., process industries adopting modular production principles).

- **Limitations (from a thesis perspective)**

EP3929679A1 is highly relevant to **orchestration**, but it still has gaps relative to your thesis goal:

- It is not inherently LLM-agentic (no native reasoning agents or dynamic tool-graph planning).
- It is not explicitly proven as vendor-agnostic in the sense required by a cross-vendor orchestrator.
- It emphasizes engineering-time orchestration; operational-time intelligence and adaptive decision-making are not the central focus.

Mini-bridge to your thesis: This patent shows how vendors treat orchestration as commercially valuable, yet it also illustrates why a next step is needed: orchestration should become intelligent, adaptive, and interoperable across heterogeneous ecosystems.

Table 4. 2 Comparison: ABB Patent vs. Research Systems

Dimension	**ABB EP3929679A1**	**ToolNet**	**AFLOW**	**SOAN**	**LLM4IAS**
Type	Patent / industrial engineering solution	Research system	Research system	Research system	Research prototype
Primary focus	Modular orchestration engineering (POL)	Tool orchestration	Workflow optimization	Self-organizing agents	Industrial integration via OPC UA
LLM-based	No	Yes	Yes	Yes	Yes
Industrial deployment	Industry-facing	No	No	No	Testbed
Vendor-agnostic	Not guaranteed (ABB-centric framing)	No	No	No	No
Module composition	Core concept	No	Partial	Strong (agent structuring)	No
Safety certification	Not specified as certified	No	No	No	No
Open source	No	Yes	Yes	Yes	Yes
Cost model	Commercial IP	Free	Free	Free	Free

The ABB patent EP3929679A1 provides a concrete and industrially validated perspective on orchestration as an explicit engineering layer, rather than an implicit by-product of low-level control logic. From the standpoint of this thesis, the most transferable insight is the conceptual separation between modular functionality and plant-level orchestration, which aligns well with the idea of treating workflows and coordination policies as first-class entities. The notion of composing systems from independently engineered modules, governed by a dedicated orchestration layer, directly informs the architectural foundations of a workflow-oriented orchestrator.

However, several core limitations prevent this solution from fully addressing the research objectives of the present work. The patent does not incorporate intelligent agent reasoning, nor does it exploit large language models for dynamic decision-making or cross-domain abstraction. Furthermore, while modular in principle, the orchestration approach remains closely coupled to vendor-specific engineering ecosystems and does not explicitly target vendor-agnostic interoperability. Consequently, although EP3929679A1 demonstrates how orchestration can be productized in industrial practice, it also highlights the absence of an intelligent, agent-driven, and vendor-neutral orchestration layer, which constitutes a central gap that this thesis aims to address.

- **4.3.2 Phaidra Patent US20250021061A1 / US12326701B2 - Deterministic AI Setpoint Governance over Legacy Control**

Patent identification.

This subsection analyzes the Phaidra patent family including US20250021061A1 (published 2025-01-16, priority 2023-07-11) and the granted US12326701B2 (as you noted in your review), with inventors including Christopher R. Vause, Frank Nerkowski, and Raahul Singh (per the patent

record). The patent targets a practical question that many industrial operators consider non-negotiable: *How can we introduce AI-driven optimization without sacrificing deterministic safety and operator trust?*

- **What the patented solution is**

Unlike many research works that optimize end-to-end control with AI, Phaidra's approach is structurally conservative and therefore commercially meaningful: it inserts an AI agent into an industrial control loop without replacing the legacy controller. Instead, the patent establishes a governance architecture that decides whether the system should follow:

- an AI-proposed setpoint, or
- a legacy controller setpoint (fallback).
- How it works (technical core supported by Figure 2)

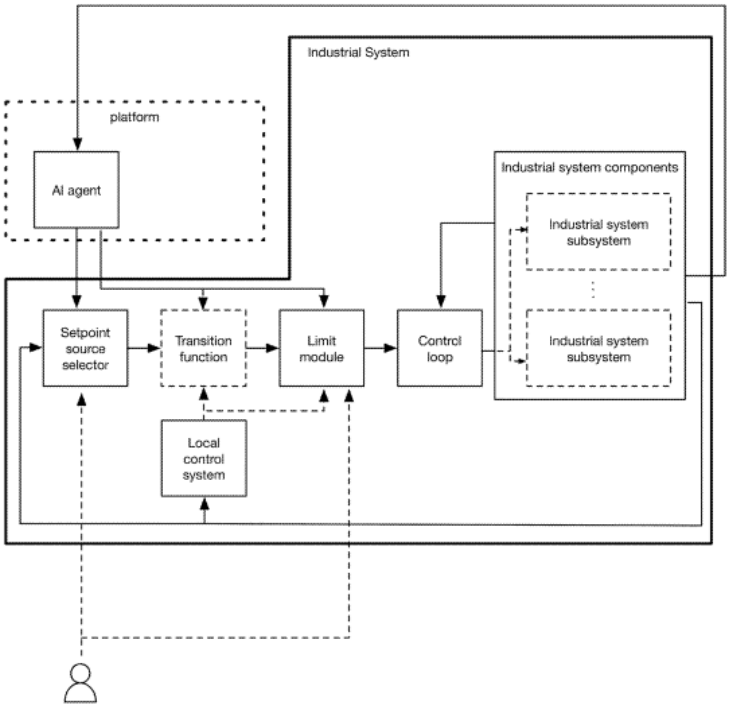


Figure 4. 2 Deterministic AI Setpoint Governance over Legacy Control

Figure 2 captures the patent's main architecture:

- An AI agent produces candidate setpoints.
- A Setpoint Source Selector chooses the setpoint source (AI vs. local control).
- Optional blocks like a transition function and limit module govern smoothness and enforce bounds.
- The resulting setpoint enters the plant's control loop, while the industrial system remains intact.

A concrete way to explain the selector behavior (as required academically) is to present it as a deterministic decision policy:

- Use AI setpoints only if:
    - the operating state is within a known envelope,
    - the AI recommendation passes constraint checks,
    - confidence/health indicators meet thresholds,
    - and there is no operator override.
- Otherwise, revert to the legacy controller setpoints.

The limit module is critical: even when AI is trusted, the output is bounded by hard constraints before it reaches the control loop. This is exactly the kind of "deployable" pattern that industrial patents often emphasize: introduce intelligence, but keep governance deterministic.

- **Why it was patented (commercial motivation)**

This architecture directly addresses adoption barriers:

- Operators and auditors require predictable fallback behavior.
- Engineers want an upgrade path that does not require re-certifying the entire control stack.

- Businesses want energy savings or performance gains, but not at the cost of operational risk.

The patent protects the design pattern that makes AI "sellable" in industrial contexts: AI as a governed advisor rather than an uncontrollable controller.

- **Who uses it (practical target users)**

This is closer to a "productizable" pattern than many academic prototypes:

- facility and energy managers who care about KPI improvements (energy, efficiency, yield),
- industrial operations teams who need safe integration with existing control,
- and control engineers who require that the original controller remains active and understandable.

- **Limitations**

From the thesis viewpoint, Phaidra's solution is valuable—but it solves a different layer of the stack:

- It is not a workflow orchestrator across heterogeneous vendor toolchains.
- It focuses on setpoint governance, not multi-agent planning and tool-graph orchestration.
- Vendor-agnostic integration is not the main claim; rather, it is about safe insertion into existing loops.

This patent is still important to you because it provides a concrete, industrially credible pattern for AgentOps-style governance (fallback, bounds, overrides). That governance mindset is exactly what your orchestrator must adopt when interacting with real industrial tools.

Table 4. 3 Comparison: Phaidra Patent vs. Research Systems

Dimension	**Phaidra (US20250021061 A1 / US12326701B2)**	**ToolNet**	**AFLOW**	**SOAN**	**LLM4IAS**

Type	Patent / deployable control governance	Research system	Research system	Research system	Research prototype
Primary focus	Safe AI setpoint selection + constraints	Tool graph orchestration	Workflow optimization	Self-organizing coordination	Industrial control integration
LLM-based	Not required (can be RL/ML)	Yes	Yes	Yes	Yes
Industrial deployment	Product-oriented	No	No	No	Testbed
Vendor-agnostic	Indirect (works over existing systems)	No	No	No	No
Safety/governance	Core contribution (fallback + limits)	No	No	No	Partial (not certified)
Real-time suitability	Designed for operational use	N/A	N/A	N/A	Supervisory latency
Open source	No	Yes	Yes	Yes	Yes
Value to your thesis	Governance pattern for AgentOps	Scaling tools	Optimizing workflows	Architecture scaling	Industrial interface layer

The Phaidra patent family (US20250021061A1 / US12326701B2) contributes a complementary but fundamentally different perspective by focusing on governance and safety-aware integration of AI into operational control loops. Its most valuable contribution to this thesis lies in the explicit separation between intelligent decision-making and deterministic execution, enforced through fallback mechanisms, constraint validation, and operator override capabilities. This design embodies a practical realization of what is increasingly referred to as *AgentOps*: ensuring that intelligent systems remain observable, controllable, and bounded in safety-critical environments.

Nevertheless, the scope of the Phaidra solution is intentionally narrow. It does not address orchestration across multiple tasks, tools, or heterogeneous systems, nor does it provide mechanisms for workflow composition, agent coordination, or cross-vendor interoperability. The

patented architecture governs *how* an AI suggestion may be accepted or rejected, but it does not define *how complex industrial workflows are planned, decomposed, or coordinated across subsystems*. As such, while the governance principles of Phaidra are directly transferable to the operational safeguards of the proposed orchestrator, the patent does not resolve the higher-level orchestration and interoperability challenges that form the core research focus of this thesis.

5. Qualitative Comparison of Existing Solutions

Building upon the analytical review of research-based systems (Section 4.2) and patent-driven industrial solutions (Section 4.3), this section formulates a structured qualitative comparison of representative solutions in the considered subject area.

The purpose of this comparison is to explicitly identify which qualitative characteristics are currently addressed by existing solutions and, more importantly, which remain insufficiently supported.

In contrast to quantitative benchmarking, this section adopts a qualitative, criteria-driven perspective, aligned with the requirements of industrial automation and intelligent workflow orchestration. The comparison is grounded exclusively in explicit evidence provided by the reviewed papers and patents, avoiding speculative or extrapolated claims.

5.1 Selection of Qualitative Characteristics for Comparison

The qualitative characteristics used for comparison were not selected arbitrarily. Instead, they were systematically derived from recurring themes, strengths, and limitations identified throughout Sections 4.2 and 4.3.

Each characteristic reflects a capability that emerged as either:

- a demonstrated strength in existing solutions, or
- a repeatedly identified limitation preventing broader industrial adoption.

Table 5. 1 Mapping of Qualitative Characteristics to Literature Findings

Qualitative Characteristic	Derived from	Justification
LLM-based reasoning	ToolNet, AFLOW, SOAN	Identified as a key enabler for high-level task interpretation, adaptive planning, and reasoning over unstructured inputs
Multi-agent coordination	SOAN, LLM4IAS	Necessary for distributed decision-making and scalable control in heterogeneous industrial systems
Workflow orchestration	ToolNet, ABB EP3929679A1	Required for structured task execution, dependency management, and reproducibility of processes
Industrial deployment readiness	LLM4IAS, ABB EP3929679A1, Phaidra US20250021061A1	Essential for applicability beyond laboratory benchmarks
Safety and certification readiness	Phaidra US20250021061A1, industrial standards	Non-negotiable requirement for compliance with IEC 61508/61511 and industrial regulations
Vendor-agnostic interoperability	Identified gap across all reviewed solutions	Critical to avoid vendor lock-in and enable cross-platform orchestration
Real-time or near real-time support	ABB EP3929679A1, industrial automation constraints	Required for supervisory and time-sensitive industrial operations
Open and extensible architecture	ToolNet, AFLOW, SOAN	Enables reproducibility, extensibility, and community-driven innovation

Notably, vendor-agnostic interoperability did not emerge as a fully addressed capability in any surveyed solution, instead appearing consistently as an unresolved gap. This observation directly informs the research motivation of this thesis.

5.2 Selection of Representative Solutions

Although the literature review comprises 25 papers, only six solutions were selected for qualitative comparison. This selection was guided by representativeness rather than quantity.

The chosen solutions satisfy at least one of the following criteria:

- They represent state-of-the-art research in LLM-based multi-agent systems with explicit architectural descriptions (ToolNet, AFLOW, SOAN).
- They demonstrate explicit validation in industrial automation contexts (LLM4IAS).
- They constitute patent-based industrial solutions addressing deployment, safety, and operational constraints (ABB EP3929679A1, Phaidra US20250021061A1).

Other reviewed works were excluded from this comparison because they:

- focused on narrower sub-problems (e.g., single-agent reasoning or non-industrial domains), or
- lacked sufficient architectural or implementation detail for systematic qualitative assessment.

5.3 Evaluation Criteria and Symbolic Scale

The qualitative evaluation follows a strict symbolic scale to ensure methodological rigor:

- **(+)** The property is explicitly implemented and supported by documented evidence (e.g., architectural descriptions, experimental results, or patent claims).
- **(0)** The property is partially addressed, conceptually mentioned, or implemented in a limited or non-scalable manner.
- **(−)** The property is not addressed, not mentioned, or explicitly excluded from the solution scope.

When ambiguity exists, the evaluation conservatively favors **(0)** over **(+)**.

5.4 Comparison of Competitive Solutions

Table 5.1 presents the systematic evaluation of six representative solutions against the eight qualitative characteristics defined in Section 5.1.

Table 5. 2 Qualitative Comparison of Existing Solutions

Solution	LLM-based reasoning	Multi-agent coordination	Workflow orchestration	Industrial deployment	Safety readiness	Vendor-agnostic	Real-time support	Open architecture
ToolNet	+	0	+	−	−	−	−	+
AFLOW	+	0	+	−	−	−	−	+
SOAN	+	+	+	−	−	−	−	+
LLM4IAS	+	+	0	0	−	−	0	+
ABB EP3929679A1	−	+	+	+	0	0	+	−
Phaidra US20250021061A1	0	−	−	+	+	0	+	−

5.5 Interpretation of Symbolic Assignments

Several partial evaluations (0) require clarification:

- ToolNet and AFLOW receive (0) in *Multi-agent coordination* because they focus on structured tool invocation rather than explicit agent-to-agent negotiation or decentralized coordination.
- LLM4IAS receives (0) in *Workflow orchestration* since its task execution relies on predefined scripts rather than dynamic graph-based orchestration.
- ABB EP3929679A1 receives (0) in *Safety readiness* because, although safety is addressed, formal certification compliance is not explicitly demonstrated within the patent documentation.
- Phaidra US20250021061A1 receives (0) in *LLM-based reasoning* due to its reliance on constrained optimization and minimal natural language interpretation.

5.6 Synthesis and Identified Research Gap

The qualitative comparison reveals a structural dichotomy between research-oriented and industrial solutions:

- Research systems (ToolNet, AFLOW, SOAN) excel in cognitive capabilities—including LLM-based reasoning, flexible workflow modeling, and open architectures—but fail to satisfy operational requirements such as deployment readiness, safety certification, and real-time guarantees.
- Industrial solutions (ABB EP3929679A1, Phaidra US20250021061A1) prioritize operational robustness, safety, and real-time execution, yet deliberately constrain cognitive autonomy, openness, and vendor interoperability.
- Hybrid systems such as LLM4IAS demonstrate the feasibility of bridging this divide but remain limited to single-vendor testbeds without formal safety validation or scalable multi-vendor support.

Quantitatively, research systems satisfy an average of 4.3 out of 8 characteristics (54%), while industrial solutions satisfy 4.0 out of 8 (50%). Critically, no analyzed solution exceeds 5 out of 8 characteristics, and vendor-agnostic interoperability remains entirely unaddressed.

This absence of solutions that simultaneously combine advanced LLM-based reasoning, multi-agent coordination, industrial deployability, and vendor-agnostic interoperability constitutes the central research gap addressed by this thesis.

6. Conclusion of the Analytical Review

This analytical review examined contemporary scientific and industrial solutions related to intelligent workflow orchestration, multi-agent systems, and the integration of large language models within industrial automation contexts. The review was conducted based on a systematic

analysis of domestic and international sources, including peer-reviewed research articles and patent-based industrial solutions published between 2022 and 2026.

The reviewed literature demonstrates a rapidly growing interest in leveraging LLM-based reasoning and agent-based architectures to enhance flexibility, adaptability, and autonomy in complex systems. Research-oriented solutions such as ToolNet, AFLOW, and SOAN highlight significant progress in cognitive capabilities, including structured workflow modeling, decentralized coordination, and advanced reasoning over heterogeneous tasks. These systems show strong potential in experimental settings and establish important methodological foundations for intelligent orchestration.

However, the analysis also reveals that such research systems remain largely disconnected from real industrial environments. Key operational requirements such as deployment readiness, safety certification, real-time constraints, and compatibility with legacy industrial infrastructure—are consistently absent or insufficiently addressed. As a result, these approaches remain confined to laboratory benchmarks and simulated workflows, limiting their immediate applicability in production systems.

In contrast, patent-based industrial solutions, exemplified by ABB EP3929679A1 and Phaidra US20250021061A1, prioritize deploy ability, robustness, and compliance with industrial safety and operational standards. These solutions demonstrate that agent-based and AI-assisted concepts can be successfully integrated into real industrial settings, particularly for supervisory control, optimization, and energy management. Nevertheless, this robustness is achieved through conservative architectural choices that significantly restrict cognitive autonomy, openness, and cross-vendor interoperability. Vendor lock-in and limited reasoning capabilities remain defining characteristics of these systems.

The qualitative comparison conducted in Section 5 confirms a fundamental structural divide between research-driven innovation and industrial realization. While academic systems emphasize intelligence and flexibility, industrial solutions emphasize stability and control. Notably, none of the analyzed solutions simultaneously satisfy all critical qualitative characteristics required for intelligent industrial workflow orchestration. In particular, vendor-agnostic interoperability emerges as a universal gap across both research and industrial domains, despite its importance for modern heterogeneous production environments.

This analytical review therefore identifies a clear and well-defined research gap: the absence of a unified framework that combines advanced LLM-based reasoning, scalable multi-agent coordination, industrial robustness, and vendor-agnostic interoperability within a single coherent architecture. Addressing this gap requires an approach that bridges the strengths of academic research with the constraints of industrial practice, rather than favoring one at the expense of the other.

The findings of this review provide a solid foundation for the subsequent stages of this thesis. They motivate the development of a vendor-agnostic multi-agent LLM workflow orchestrator designed to operate at the supervisory level, integrate with existing industrial standards, and support heterogeneous vendor ecosystems while maintaining controlled and explainable agent behavior. The next phase of the work will build directly upon these insights by formulating design requirements and proposing an architectural concept that responds explicitly to the identified limitations of existing solutions.

Chapter 1 INSERT CHAPTER TITLE

Chapter 2 INSERT CHAPTER TITLE

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